



2002 Network Tariff Report

Produced pursuant to clause 11.3.5 of Volume I of the Office of the Regulator General's Electricity Distribution Price Determination 2001 – 2005, and supplementary advisory notes

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A Reid
Manager Network Pricing
United Energy Distribution
PO Box 1185
Moorabbin Victoria 3189
areid@ue.com.au

THE PURPOSE OF THIS REPORT

This report is intended to provide guidance for customers, describing issues, principles and rationale used by United Energy in setting its distribution tariffs for the year 2002. While extensive consultation continued over the past year, comment is invited from those interested parties wishing to air their views regarding future tariff reviews and on-going development of distribution initiatives. Over the balance of the current 5-year regulatory period, United Energy proposes to further develop its pricing initiatives, with particular emphasis on signals for encouraging the efficient and equitable use of the electricity distribution network while minimising environmental impact.

It is intended that the *2001 Tariff Report* be read as a companion document to this report.

1. BACKGROUND

During 2001 the trend of peak demand increasing at a faster rate than energy usage continued. Slowing down this trend remains a key

aim in order to minimise the network service cost to customers over time.

During 2001 several initiatives aimed at encouraging customers to reduce peak demand were introduced. Uptake was however slower than expected. It is evident that customers will need a little more time to respond than anticipated. The slow uptake is thought to be due to several factors including:

- Retailers having a lack of time to build the features into their retail product offerings,
- The delay in FRC (full retail contestability), and
- In some cases the strength of the 'signal' may have been insufficient to make it worthwhile for customers to respond.

With the advent of FRC it is envisaged that Electricity Retailers will take the opportunity to promote the features of the 2002 distribution tariffs as means of helping their customers minimise costs.

2. SAVINGS FROM CUSTOMER CHOICE

Overall, United Energy has sought to increase the choices open to customers. Each customer will have their own view of the value they

place upon being able to use the network at any time versus reducing their demand at key times in return for savings. If customers feel that they forgo little by deferring their electricity usage at a few key times, they stand to make cost savings. If conversely they see real value in being able to use electricity at any time, then this is likely to come at a higher price, reflecting the higher relative cost of providing service to the 'continuous anytime' customer. As discussed in the *2001 Tariff report*, the predominance of ordinary-type (simple) meters is a hindrance to giving customers more tariff choices and enabling them to make savings by their own actions. To overcome this, UE has continued to introduce further tariffs based upon using interval-type (smart) meters. At half-hour intervals, smart meters record the kWh (energy) used over the previous 30 minutes. This information is then used to calculate customer bills based on the different rates applying to the different time zones. These zones are comprised of time of day, day of week and season. In this way, a more effective measure of each individual customer's contribution to how big the network needs to be is determined. This enables the supplier to better assess whether the customer is avoiding the high-cost-to-supply periods and to pass savings on accordingly.

3. THE EFFECT OF AIR-CONDITIONING LOADS

As also discussed in last year's report, much of the rapid increase in maximum network burden that Victorian networks have been experiencing in recent years can probably be attributed to the increasing market presence and usage of air-conditioners, particularly at the residential level.

For the majority of the year, air-conditioners can be used without needing to make the network bigger to cope with them. The main adverse impact on the network occurs, not surprisingly, when the air-conditioners are used heavily on the few days and hours of summer. This is where all the loads added together, set the size the network has to be for whole year. Customers can gain much of the benefit of having an air-conditioner but significantly reduce the network burden by running them before the peak period to pre-cool the space they are serving, and then turn them off for 3 hours at 3pm on hot summer workdays. If this is not feasible then savings can still be made by setting the thermostat to a higher temperature for the period or perhaps using a timer to run for say 10 minutes every half-hour. UE's new 2002 'hot' variants and 'reverse-cycle air-conditioning' tariffs also use the interval meters mentioned earlier. Compared to the base tariffs, the strength of the peak signal is higher

and the number of hours in the target time zone is smaller and so may be easier for customers to avoid and therefore make savings.

4. THE RELATIONSHIP BETWEEN CLIMATIC TEMPERATURE AND NETWORK LOADING

United Energy has studied climatic temperature trends in Melbourne using data from the last 30 years. This was correlated to electricity demand in recent years in order to forecast the probability of high temperature events giving rise to peak electricity demand 'events'.

Chart 1 indicates the relationship, for various broad customer segments, between their average daily demand profile on a 'peak' high temperature summer day (solid line) compared to their profile on a comparable 'normal' average temperature summer day (dotted line). The difference between the two lines is an indicator of the amount of temperature-related loads (such as air-conditioning). It appears that the residential sector has the largest proportion of

temperature-related loads but that these peak at a time later than that of the total system.

The rectangles indicate the time of day and demand band within which each segment is likely to experience its own peaks. The 'total network' peak zone is the one for which the network is generally sized and corresponds to United Energy's Summer Demand Incentive Charge (SDIC) pricing zone. Thus the respective contributions to the maximum by each group to that rectangle is an indicator of the burden they place on the system. It should be noted however that these profiles are the equivalent contribution at the 'boundary level' of the network, which is the interface between the distribution network and the bulk transmission system. In assessing the costs of service to each sector, account must also be taken of the level of connection and amount of network assets required to provide supply. Table 1 provides further detail on which segments use which network asset categories.

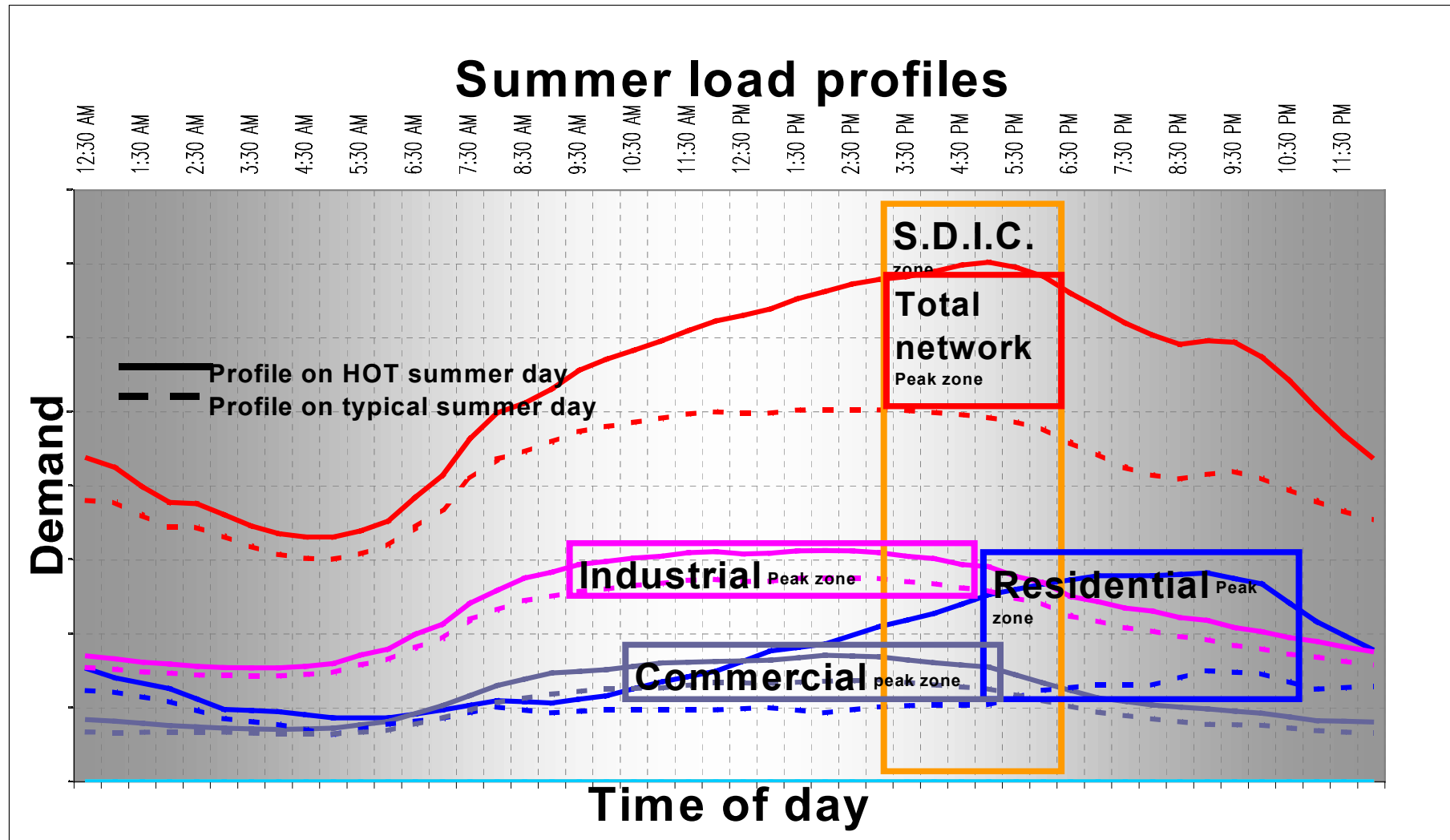


Chart 1 Summer load profiles

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Table 1 ‘Asset participation matrix’

5. CONSULTATION WITH CUSTOMERS

Feedback from larger customer groups on the whole supported UE's increased levels of time of use signals but there was some concern that the signals were not 'sharp' enough for it to be feasible or worthwhile for customers respond. Using the climatic study results, a set of new tariffs was developed based on focussing the signals on to only those hot summer days that are very likely to contribute to increasing the annual peak network demand.

6. NEW TARIFFS¹

6.1 'HOT' variant tariffs

- **HOT LVkW TOU** (*HOT Low voltage kilowatt time of use*)
- **HOT kVA TOU** (*HOT Low voltage kilovolt-Amp time of use*)
- **HOT HV kVA TOU** (*HOT High voltage kilovolt-Amp time of use*)

The new tariffs are variants of existing tariffs. The base tariff can be identified by removing the 'HOT' prefix from the above tariff names.

¹ Some conditions apply

The variants and their base tariffs are the same as each other in most respects except that the 'summer demand incentive charge' (SDIC) is nearly double the normal rate in the variants, but only applies if the maximum daily ambient temperature is forecast to reach or exceed 30°C. The normal SDIC rates apply to all summer workdays, whereas the hot version may only be triggered by perhaps 5 to 20 days per summer. In a given summer month, there is also a chance that no days end up with forecasts over 30°C, in which case no SDIC charge would accrue for that month. This, combined with providing a stronger signal that is easier to respond to, is largely the reason for higher rate. Thus on the HOT tariffs, the full benefit of responding to the SDIC can be gained by customers responding on only a small number of days over the five months of summer.

6.2 RCAC kW TOU tariff

(Reverse-cycle air-conditioning kilowatt time of use)

In addition to providing summer cooling, a 'reverse-cycle' type of air-conditioner has the added customer benefit of providing a very efficient source of heating during colder periods of the year. This also provides an improved use of capital as near year-round usage is possible.

The RCACkWTOU tariff is available only to those customers that have above a minimum amount of reverse-cycle air-conditioning capacity installed. This give access to low energy rates at off-peak times, offset with a relatively high energy rate at peak times, and a high demand charge for the 'SDIC' hours (3 to 6 pm summer workdays). The attractiveness of this tariff will depend upon the customer's willingness and ability to avoid peak summer usage at the key times. The low running cost during winter should encourage usage over less efficient or environmentally-friendly heating alternatives. The network has some unutilised capacity during the winter because with colder ambient temperatures, the network's rating generally increases as well as the combined loads of other customers being less than peak times in the summer.

This tariff is only available if an 'interval' (kW) meter is installed.

There are a number of conditions including a minimum appliance capacity installed.

6.3 W.E.T. tariff (*Winter Economy Tariff*)

The WET tariff is available only to those customers with above a threshold capacity of permanently-wired electric panel heaters installed. The tariff is designed to enable those customers who are currently on the LVS1R tariff who have a 'simple' meter to also

gain access to some benefit from time of use pricing signals by switching to this new tariff. The usage in the non-summer exceeding 1,020 kWh/quarter, has a rate less than half the normal rate.

There are a number of conditions including a minimum appliance capacity installed.

7. DISCONTINUED TARIFFS

In designing its 2002 distribution tariffs United Energy has sought to build on the initiatives introduced in 2001, and incorporate additional feedback from customers. This has resulted in discontinuation of the 'interruptible' tariff variants.

The interruptible tariffs proved to have insufficiently strong signals to be attractive to customers. Customers were also not willing to provide sufficient 'surety' that they would definitely activate load reduction at critical times. From the network point of view, unless reductions can be relied upon, the network capacity needs to be planned as if the full load is still present in order to avoid undue impacts on other customers. While the strength of the signals may have been increased, this may have lead to economically inefficient

outcomes due to over-signalling beyond the value created in avoiding network costs.

The introduction of the 'Hot' tariffs is intended to provide an alternative to customers who found the concept behind the interruptible tariff attractive, but were unwilling to delegate some 'control' to UE's control room.

8. TARIFF STRUCTURE

Changes to 2001 tariff structures have been kept to a minimum. In response to customer feedback, the main change applies to those tariffs incorporating the 'SDIC' (Summer Demand Incentive Charge). For these tariffs, the minimum demand quantity for the SDIC has been reduced to zero, but minimums still apply to the rolling average demand charge. This should avoid the previous situation where below the minimum SDIC level there was little incentive to further reduce demand. Given that not all customers are on these tariffs or may not be able to respond, there is generally a benefit to the network at peak times in reducing demand as much as possible.

9. TARIFF RATES

Changes to 2001 tariff rates have been kept to a minimum, but do seek to incorporate customer feedback in minimising some of the unintended effects that last year's widespread changes gave rise to. There has been some re-balancing within categories in order to increase the emphasis on demand-based charges over energy-based charges. The changes also include some incremental 're-balancing' of rates between customer groups in light of year-to-year trends in practice. On the tariffs where the standing charge still applies – those without an interval meter – the rates have been uniformly increased to reflect costs and inflation effects. For interval meter-based tariffs, there is no standing charge because the presence of demand-based charges (i.e. SDIC & rolling peak demand charges) tends to lessen the need for a fixed charge to cover fixed costs.

10. DISTRIBUTION-CONNECTED GENERATORS

The presence of distributed generator customers amongst diversified load customers can have a beneficial effect on the investment in electricity networks.

A tariff suitable for embedded generators is being developed and is under discussion with regulatory authorities. It will have the objective of ensuring that embedded or distributed generators be

appropriately rewarded for avoided transmission costs, and also bear their share of distribution costs with other users of the network rather than becoming effective 'free-riders' on the network that load-based customers pay for. It seems more equitable that load customers have charges reduced if there are other parties gaining value from being connected to the network and that those other parties faces common charges - as loads have to.

This could be done in a manner that does not increase total revenue for the network, rather more fairly and equitably sharing the costs of the network amongst all users. At present, the regulatory authorities appear reluctant to enable the reduction of charges to load customers that would be possible if a charge were applied to customers whose main business is generation. All customers who are connecting generators are being made aware that charges for their common use of the network may apply in future.

11. ALLOCATION OF TRANSMISSION COSTS

Under the regulatory mechanism governing the pass-through of transmissions costs to distribution customers, the allowed amount may be different to the amount actually accruing or collected for that

year. The differences are accounted for under the 'k-factor' mechanism.

Due to favourable outcomes on transmission costs for 2001, such as 'residue settlements', transmission revenue recovered exceeded transmission costs affecting the 'k-factor'. The TUOS rates have been reduced in order to lessen the amount collected in 2002.

United Energy is concerned that the current 'k-factor' mechanism can produce substantial year-to-year volatility. On the one hand this produces flow-on volatility to customers and on the other exposes distributors to a lack of revenue symmetry. The Transmission pricing changes anticipated as a result of the recent ACCC authorisation of changes to the National Electricity Code, may further exacerbate this problem for Distributors in 2003.

A modified mechanism, used in other parts of the world, would permit this volatility to be smoothed out, improving certainty for all parties.

12. UPPER & LOWER BOUNDS ON AVERAGE COST (average cents per kWh)

These upper and lower bounds represent two forms of economic test on the levels of tariffs and provide some guidance as to whether

tariff rates are too high or too low to be sustainable. The upper and lower bounds were derived for the 2001 Tariff report and have not been re-calculated for this report - other than to account for inflation effects. It can be seen that each tariff category has at least one tariff option that is in the mid-range between the two limits.

13. CHANGES IN AVERAGE ANNUAL CHARGES TO EACH CUSTOMER TARIFF MARKET SEGMENT

Residential single and dedicated rates have remained virtually unchanged in terms of average cents per kWh over the year.

Some re-balancing has taken place lowering average rates to larger customers and generally increasing rates in tariffs that are closed to new connections in order to provide a greater incentive for existing customers to consider transferring to the newer tariffs that have signals for more efficient consumption.

Table 2 provides more detail on changes in average rates over an annual period.

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Table 2. Estimated change in effective tariffs (average annual effective cents/kWh across tariff class)

Tariff group	2001 Tariff (NUOS excl GST)	2002 Tariff (NUOS excl GST)		Lower bound	Upper bound
	Average annual cost (cents/kilowatt hour)	Average annual cost (cents/kilowatt hour)	% difference	“Avoidable cost” (cents/kilowatt hour)	“Stand-alone cost” (cents/kilowatt hour)
LVS1R	5.60	5.59	0.0	0.6	10.2
LVS2R*	3.70	3.76	1.6		
LV Ded*	1.36	1.36	0.0		
W.E.T. <i>(new)</i>		4.80			
RCAC <i>(new)</i>		3.38			
LVkWTOU	5.27	5.07	<i>(-3.7)</i>	0.6	9.3
LVM1R	7.15	7.26	1.5		
LVM2R5D*	5.88	5.83	<i>(-0.7)</i>		
LVM2R7D*	6.90	7.03	1.8	1.8	22.8
LV Unmetered	2.97	3.05	2.7		
LV1R*	4.94	4.93	<i>(-0.3)</i>	0.5	5.1
LV2R*	5.22	5.14	<i>(-1.6)</i>		
LVkVATOU	3.29	3.19	<i>(-2.8)</i>		
HvkVA	1.82	1.77	<i>(-3.0)</i>	0.4	3.2
* these tariffs are closed to new connections					

14. THE FUTURE DIRECTION OF UNITED ENERGY'S TARIFFS

United Energy remains committed to seeking beneficial outcomes by providing choice and incentives to customers via innovation in its distribution tariffs. In the last two years UE has put in place tariffs that address many of the major areas of opportunity for stimulating market responses in order to reduce costs. Much of this has consisted of providing economic signals to consumers as to the main drivers of costs in providing an electricity delivery network. Given that a wider choice of effective tariff options is now offered, the most important decisions are in the hands of the customer

- respond and make savings, or
- not respond, but realising that the costs of service will be higher, in relative terms, if the latter is chosen.

Electricity retailers have a key role to play in 're-packaging' distribution and transmission cost signals into 'retail' products. The more effective range of distribution tariffs now available offers an increasingly flexible platform of opportunities to retailers, whether they be mass or niche-market orientated.

While full-retail contestability has only recently been launched, this may stimulate an increased level of activity based on responding to economic signals provided by networks.

It is evident that in order for demand-side investments to be attractive, distribution pricing signals and structures need to remain reasonably stable over time so that there is time for them to produce a worthwhile return. While UE cannot with certainty forecast the regulatory climate beyond the current 5-year period, it will endeavour to provide as steady a climate as possible for DSM investment by its customers.

Specifically, UE sees the following as key aspects of its tariff programme going forward;

- Encouraging an increased uptake of interval-metering based tariffs such as the current 'TOU' series;
- Encouraging customer demand-side initiatives
- Preserving as far as possible a stable commercial climate for investment in DSM equipment or practices

- The SDIC concept will remain, but the time window may be updated from time to time in order that it remain aligned with the key network peak demands
- Cost-of-supply modelling will update annually to reflect changes in relative contributions from segments
- Further accentuating the emphasis on peak season (summer), day of week and time of day in order to stimulate DSM response
- Properly integrate the contribution that distribution-connected generator customers should be making to the costs of providing network services that all users share and derive value from.

15. LET US KNOW YOUR VIEWS

United Energy continues to encourage customers to discuss ideas on how mutual needs may be better met. Many of the 2002 changes will be familiar to those customers that participated in various workshops, seminars and on-on-one sessions hosted by UE during 2001. It is hoped that this can be further enhanced during the next year.

Appendix 1

The distribution (DUOS) and transmission (TUOS) tariffs for 2002

Schedule of **Network** Use of System tariffs (**NUOS**) from 1.1. 2002, excluding GST

Schedule of **Distribution** Use of System tariffs (**DUOS**) from 1.1. 2002, excluding GST

Schedule of **Transmission** Use of System tariffs (**TUOS**) from 1.1. 2002, excluding GST

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NETWORK tariffs (Distribution + Transmission)

Schedule of Network (Distribution+Transmission) Use of System (NUOS) Tariffs: 1 January 2002									
Network Tariff Component									
<u>Rates Exclude GST</u>	Standing Charge	Summer Peak Energy Charge	Non-Summer Peak Energy Charge	Off Peak Energy Charge	Rolling Peak Demand Charge	Summer Demand Incentive Charge (SDIC)	Eligibility (demand)	Eligibility (consumption)	Minimum Chargeable Rolling Demand
Tariff Name	\$/cust/yr	c/kW h	c/kW h	c/kW h	\$/kVA/yr	\$/kW/mnth or \$/kVA/mnth	kVA	MW h (000's kW h)	kVA
LVS1R	25.534	5.671	4.536	-	-	-	Residential customer		-
WET-Step 1<= 1,020/90days	25.532	5.671	4.536	-	-	-	0-20	0-70	-
-Step 2 > 1,020/90days	-	-	2.117	-	-	-	0-20	0-70	-
RCACKW TOU	-	12.519	2.928	1.356	-	18.529	0-20	0-70	-
LVS2R*	36.796	8.056	6.445	1.356	-	-	-	-	-
LVDed*	-	-	-	1.356	-	-	-	-	-
LVM1R	33.192	7.780	6.441	-	-	-	Business customer		-
LVM2R5D*	47.835	9.291	7.626	1.356	-	-	-	-	-
LVM2R7D*	47.835	9.229	7.614	1.356	-	-	-	-	-
LVUnmetered	-	8.150	6.365	1.356	-	-	-	-	-
LVL2R*	47.835	8.189	6.710	1.356	-	-	-	-	-
LVL1R*	33.192	5.422	4.488	-	-	-	-	-	-
LVkW TOU	-	6.317	5.053	1.356	-	8.873	0-150	0-400	-
HOT LVkW TOU	-	6.317	5.053	1.356	-	17.053	0-150	0-400	-
LVkVATOU	-	1.312	1.050	0.895	47.305	4.683	150 +	400 +	150
HOT LVkVATOU	-	1.312	1.049	0.895	47.303	9.002	150 +	400 +	150
HVkVATOU	-	0.846	0.676	0.579	32.460	3.174	150 +	400 +	1,150
HOT HVkVATOU	-	0.846	0.676	0.579	32.460	6.100	150 +	400 +	1,150
SubTkVATOU	-	0.472	0.378	0.257	6.048	0.633	150 +	400 +	11,100
S (Small), M (Medium), L (Large), LV (Low Voltage), HV (High Voltage), SubT (Subtransmission), TOU (Time of Use), R (rate), D (Day)									
* indicates tariff is closed to new connections. New connections requiring off peak network charge rates must connect to one of the Time of Use (TOU) rates and install an interval meter.									

DISTRIBUTION tariffs

Schedule of Distribution Use of System (DUOS) Tariffs: 1 January 2002									
Network Tariff Component									
Rates Exclude GST	Standing Charge	Summer Peak Energy Charge	Non-Summer Peak Energy Charge	Off Peak Energy Charge	Rolling Peak Demand Charge	Summer Demand Incentive Charge (SDIC)	Eligibility (demand)	Eligibility (consumption)	Minimum Chargeable Rolling Demand
Tariff Name	\$/cust/yr	c/kWh	c/kWh	c/kWh	\$/kVA/yr	\$/kW/mnth or \$/kVA/mnth	kVA	MWh (000's kWh)	kVA
LVS1R	25.534	4.894	3.914	-	-	-	Residential customer		-
WET-Step 1<= 1,020/90days	25.532	4.894	3.914	-	-	-	0-20	0-70	-
-Step 2 > 1,020/90days	-	-	1.495	-	-	-	0-20	0-70	-
RCACKWTOU	-	4.476	1.138	1.356	-	16.989	0-20	0-70	-
LVS2R*	36.796	6.958	5.566	1.356	-	-	-	-	-
LVDed*	-	-	-	1.356	-	-	-	-	-
LVM1R	33.192	6.881	5.722	-	-	-	Business customer		-
LVM2R5D*	47.835	8.734	7.180	1.356	-	-	-	-	-
LVM2R7D*	47.835	8.377	6.932	1.356	-	-	-	-	-
LVUnmetered	-	7.251	5.646	1.356	-	-	-	-	-
LVL2R*	47.835	7.632	6.264	1.356	-	-	-	-	-
LVL1R*	33.192	4.523	3.769	-	-	-	-	-	-
LVkWTOU	-	5.769	4.615	1.356	-	8.103	0-150	0-400	-
HOT LVkWTOU	-	5.769	4.615	1.356	-	15.573	0-150	0-400	-
LVkVATOU	-	0.967	0.774	0.895	42.183	4.104	150 +	400 +	150
HOT LVkVATOU	-	0.967	0.773	0.895	42.181	7.888	150 +	400 +	150
HVkVATOU	-	0.626	0.500	0.579	27.130	2.711	150 +	400 +	1,150
HOT HVkVATOU	-	0.626	0.500	0.579	27.130	5.210	150 +	400 +	1,150
SubTkVATOU	-	0.324	0.259	0.257	2.230	0.250	150 +	400 +	11,100
S (Small), M (Medium), L (Large), LV (Low Voltage), HV (High Voltage), SubT (Subtransmission), TOU (Time of Use), R (rate), D (Day)									
* indicates tariff is closed to new connections. New connections requiring off peak network charge rates must connect to one of the Time of Use (TOU) rates and install an interval meter.									

TRANSMISSION tariffs

Schedule of Transmission Use of System (TUOS) Tariffs: 1 January 2002									
Network Tariff Component									
Rates Exclude GST	Standing Charge	Summer Peak Energy Charge	Non-Summer Peak Energy Charge	Off Peak Energy Charge	Rolling Peak Demand Charge	Summer Demand Incentive Charge (SDIC)	Eligibility (demand)	Eligibility (consumption)	Minimum Chargeable Rolling Demand
Tariff Name	\$/cust/yr	c/kWh	c/kWh	c/kWh	\$/kVA/yr	\$/kW/mnth or \$/kVA/mnth	kVA	MWh (000's kWh)	kVA
LVS1R	-	0.777	0.622	-	-	-	Residential customer		-
WET-Step 1<= 1,020/90days	-	0.777	0.622	-	-	-	0-20	0-70	-
-Step 2 > 1,020/90days	-	-	0.622	-	-	-	0-20	0-70	-
RCACKWTOU	-	8.043	1.790	-	-	1.540	0-20	0-70	-
LVS2R*	-	1.098	0.879	-	-	-	-	-	-
LVDed*	-	-	-	-	-	-	-	-	-
LVM1R	-	0.899	0.719	-	-	-	Business customer		-
LVM2R5D*	-	0.557	0.446	-	-	-	-	-	-
LVM2R7D*	-	0.852	0.682	-	-	-	-	-	-
LVUnmetered	-	0.899	0.719	-	-	-	-	-	-
LVL2R*	-	0.557	0.446	-	-	-	-	-	-
LVL1R*	-	0.899	0.719	-	-	-	-	-	-
LVkWTOU	-	0.548	0.438	-	-	0.770	0-150	0-400	-
HOT LVkWTOU	-	0.548	0.438	-	-	1.480	0-150	0-400	-
LVkVATOU	-	0.345	0.276	-	5.122	0.579	150 +	400 +	150
HOT LVkVATOU	-	0.345	0.276	-	5.122	1.114	150 +	400 +	150
HVkVATOU	-	0.220	0.176	-	5.330	0.463	150 +	400 +	1,150
HOT HVkVATOU	-	0.220	0.176	-	5.330	0.890	150 +	400 +	1,150
SubTkVATOU	-	0.148	0.119	-	3.818	0.383	150 +	400 +	11,100
S (Small), M (Medium), L (Large), LV (Low Voltage), HV (High Voltage), SubT (Subtransmission), TOU (Time of Use), R (rate), D (Day)									
* indicates tariff is closed to new connections. New connections requiring off peak network charge rates must connect to one of the Time of Use (TOU) rates and install an interval meter.									

Appendix 2

Terms used:

DSM	Demand-side management. Refers to the ability of customers that consume a product to change behaviours that alter the overall balance between supply and demand. Examples may include the installation of energy management systems, or deferring consumption at time of peak demand.
‘free-rider’	A party that gains an economic benefit from something invested in by another party.
Load factor	Load factor is a measure of how ‘steady’ a load is. It is the extent to which average usage compares to maximum usage. It is generally much more efficient to supply a high load factor (100%=constant unchanging demand) than a ‘peaky’ (less than 100%) usage pattern. This is because depending upon the time of the peak, the capacity needs to be there all the

time to supply the peak, even though most of the time it may not be fully required.

LVS1R etc Abbreviated Tariff names.

- The 1st two letters refer to the voltage level at which the customer is supplied, **LV** = low voltage 240V(singleφ) or 415V(threeφ), **HV** = high voltage 6.6, 11 or 22kV, **ST** = sub-transmission 22 and 66kV. **HOT** = tariff variant based on high ambient temperature threshold
- The 2nd section refers to the type of customer, **S** = small, **M** = medium, **L** = large, no letters generally indicates ‘generic’ interval metered customer.
- 3rd section indicates type of rate structure, **1R** = single energy rate, **2R** = peak/off-peak rate, **kW** =

	interval metered in kW, kVA = interval metered in kVA .		A regulatory method aimed at ensuring differences between ‘actual’ and ‘forecasted’ are washed-up year to year on a revenue neutral basis. This applies to Transmission wholesale costs and TUOS revenues.
kW	Kilowatts , (1,000 Watts) the ISO unit of power measurement		
kWh	An I.S.O. unit used for the measurement of energy. It is the product of kW times time (in hours).	ORG	Office of the Regulator-General of Victoria now Essential Services Commission
kVA, and power factor	kilo-Volt-Ampere , the vector product of voltage times current. It is a measure of the proportion of useful (kW) power delivered vs. the current (Amps) drawn by the load, this is known as ‘power factor’. The size of assets required is generally a function of the current. Therefore the closer the power factor is to 1.00, the smaller the asset required to supply the load.	Rate shock	The change in average price to a customer caused by changes in price structure or level, compared to prices prior to the changes.
k-factor mechanism		Residue settlement	A means by which over-recoveries of costs on the part of the energy and transmission markets are rebated back to customers.
		‘Revenue symmetry’	Refers in this context to the extent to which any ‘unders’ and ‘overs’ are corrected in an identical

manner. If say under-recoveries are capped at 2% but over-recoveries are passed on in full at say 20%, this leads to an unsymmetrical outcome in that it could take 10 years to get back to neutrality.

SDIC Summer Demand Incentive Charge

Supply-side Similar to **demand-side** described earlier, but this time referring to **providers** of services in the customer supply chain. In the case of this paper, the term refers specifically to electricity distributors and generally to the industry including transmission and generation.

Tariff Basket The form of the regulatory Distribution price control in effect since 1 January 2001. Distribution revenue is the same as tariff revenue. The regulated revenue amount is the same as actual tariff revenue and the "correction factors" now apply only to transmission revenue.